

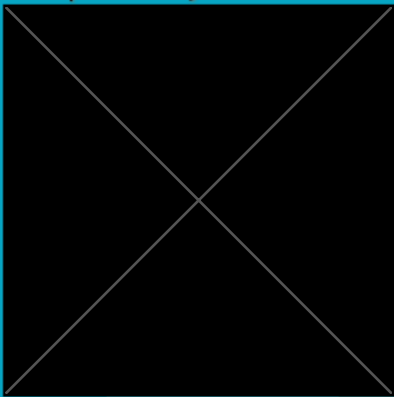
**VICTORIA
UNIVERSITY**

NIT3003 IT Capstone Project 1
Block 4, Semester 1 2025
Footscray Park

PROJECT PROPOSAL :

Enhancing Flood Risk
Detection through Localised
Weather Data Analysis

Prepared by:



Submission Date:

16th June 2025

Table of Contents

Introduction.....	4
General Background.....	4
Market Analysis.....	4
Competitor Analysis: An Overview.....	5
Project Aims and Unique Selling Proposition.....	6
Purpose and Scope.....	7
Functional Requirements.....	8
Environmental Sensor Integration	8
Data Preprocessing	8
Data Storage and Management	8
Predictive Analytics	8
Data Visualisation and Reporting	8
Alerting and Notification Systems	9
Non-functional Requirements.....	10
System reliability	10
Uptime monitoring	10
System redundancy	10
API support	10
Scalability	10
Use Case.....	11
Extreme Weather Event Notification	11
Generating Risk Report	11
Configuring Alert Thresholds	13
Deploying Environmental Sensor.....	14
Resource Management.....	15
Introduction	15
Identification of Required Resources	15
Allocation of Resources	16
Resource Scheduling.....	17
Contingency Planning.....	17
Monitoring.....	17
Evaluation	18
Sequence Diagram.....	19
UML Source Code.....	20

Pseudocode	21
Data Preprocessing Logic.....	21
Predictive Model Threshold Evaluation	21
Alert Notification Triggers.....	21
Data Query via Web Application Dashboard	22
UI Design	23
Dashboard	23
Login Page.....	24
Risk Management Plan	25
Risk Assessment	25
Risk Mitigation.....	26
Timeline	28
NIT3003 – IT Capstone Project 1	28
Project Implementation	29
NIT3004 – IT Capstone Project 2	29
Conclusion	30
References	31

Introduction

General Background

AI which stands for artificial intelligence, is a field within computer science which heavily focuses on creating systems that can perform tasks requiring human intelligence. These tasks can be a range of things such as learning from data, recognizing patterns, making decisions and adapting to dynamic environments. AI has evolved at a very fast rate with advances in machine learning, neural networks, and data processing. In today's world, AI has the possibility to power everything from personal assistants to real-time flood prediction systems.

Recently, AI has been garnering momentum as a distributive technology and has been applied in various domains around the world. One of them being environmental monitoring and urban infrastructure. AI's ability to handle huge volumes of data and recognise patterns and predictions has made it a relevant tool to enhance efficiency and perceptiveness of city systems. AI can process real-time weather data collected by multiple sensors to predict system overloads and provide for emergency responses. The scope of applying AI in the local infrastructure forms a big contribution in building resilient data-driven cities.

Urbanization, climate change, and aging infrastructure are coalescing to cause problems for many city councils and urban planners when it comes to managing stormwater and migrating flood risk. Melbourne has been more susceptible to extreme weather patterns over recent years, particularly heavy rain events which can overload current drainage infrastructure. These events are likely to become more frequent, and with climate changing variability, stormwater management is an increasingly urgent investment.

In response to this, local governments are turning towards smart, data-driven solutions so that they can monitor, predict and respond to flood risks in real-time. One of the most common approaches now involves IoT weather sensors which are able to collect local environmental data such as rainfall, wind speed, humidity and temperature. When deployed correctly, these network sensors can enhance stormwater management by feeding data into an AI-based predictive model which can forecast flood risks.

The council is planning the deployment of new stormwater sensors across the Maribyrnong area which are designed to monitor and collect water flow data on their drainage infrastructure. To optimise the effectiveness of these sensors, there is a unique opportunity to complement them with the use of AI and machine learning algorithms. This can provide considerable benefits to forecasting accuracy if local weather data is taken into account; such information may be ignored by conventional systems. AI will collaborate with these sensors to detect and anticipate overflowing or flood situations, thereby improving the security and safety at a community level.

Market Analysis

The overall market analysis for smart environmental monitoring and flood management technology is currently experiencing significant growth due to the introduction of seamless AI solutions becoming more affordable to consumers. The specific customer we are here to provide for is the Footscray municipal council as the frequency of extreme weather events, rising population and growing concerns surrounding flooding and disaster prevention techniques increases so does the demand for a smart solution to be introduced. The ultimate step is to provide the integration of IoT and AI into municipal

infrastructure as an emerging strategic priority for the local community to allow for maximal efficiency and results.

According to the AEM blog section under weather station sensors '[Weather Station Sensors: Types, Use Cases, & Selection Criteria](#)' AEM goes into heavy detail on which type of sensors they use and why they use them. AEM get their sensors through companies like High Sierra Electronics and Lambrecht Meteo.

The market for smart environmental monitoring and flood management technology is experiencing a strong and substantial growth. This is mainly due to the increasing demand for real-time data, predictive analytics and automated response systems. A key factor for this growth is the expanding accessibility of AI and IoT technologies becoming more affordable and scalable across infrastructure projects.

The Maribyrnong City Council is an ideal place for such an innovative solution to take advantage of it. With its aging drainage infrastructure, increasing population and the growing concern of climate change; the need for an intelligent and adaptive flood management solution becomes urgent. As storm events become less predictable, the limitations of traditional flood response systems become more apparent. This highlights the countless opportunities for the implementation of real-time, data driven alternative systems.

In this project, we aim to respond to that opportunity by integrating an AI-powered predictive system and an IoT-based sensor network into Footscray's drainage ecosystem. Such a solution is already aligned with local strategic goals outlined in the Council's Integrated Water Management Plan (2023-2033) and the Flood Management Plan (2016-2021).

Competitor Analysis: An Overview



There is a wide range of existing, well-established companies currently dominating the Melbourne AI-enhanced flood monitoring systems market. However, there are still gaps to be filled with these business types. These include predictive performance, integration, cost, and community participation. Our system is well placed to perform best in terms of these organizations, and to overcome their limitations with its next-generation AI, open data interfaces, low-cost hardware, and easy to use interface.

Competitor 1 - EWS Australia <https://ewsmonitoring.com/service/flood-monitoring/>

EWS provides a robust flood monitoring system, which uses hardware and real-time data alerts. Their focus is to provide early warnings through fast data transmission via SMS or email.

Strengths:

- Reliable and durable hardware which is suited for harsh environments
- Real time alerts which is available across Australia
- Minimal data latency

Weaknesses:

- Lacks advanced AI and predictive modelling capabilities
- Has limited integration with third-party platforms or smart city infrastructure
- Has a heavy focus on hardware, with very little software intelligence

Competitor 2 - LiXiA <https://lixia.com.au/flood-disaster-monitoring>

LiXiA offers flood monitoring that is highly accurate and tailored towards local councils, often outperforming the Bureau of Meteorology in detecting flood peaks.

Strengths:

- Very fast flood peak detection
- Designed for council use with situational dashboards

Weaknesses:

- Limited information on their AI use and their long-term learning capability
- Primarily focused on water level detection, without broader factors in mind such as air temperature, wind speeds, and humidity.

Competitor 3 - Avant <https://www.avantglobal.ai/water-level-monitoring-and-alerts>

Avant uses AI and digital twin modelling so that they can simulate real-time and future flood events. Their platform can use multiple data streams for advanced scenario planning.

Strengths:

- Strong integration of AI and digital twin technologies.
- Has a customisable platform which is built for large scale infrastructure modelling
- Predictive forecasting under various environmental scenarios.

Weaknesses

- High cost and complexity, so could be inaccessible for smaller councils
- Their integration demands significant resources and data
- Involves a steeper learning curve and longer implementation timeline

Project Aims and Unique Selling Proposition



The main aim of this project is to enhance Maribyrnong City Council's stormwater management capabilities which will be done by deploying stormwater sensors across strategic locations. When these sensors are integrated with the AI, this data will enable for early flood warnings to be sent out, targeted maintenance, and improved decision making.

With the AI being able to predict flood events with the data from the sensors, alerts can be sent to local councils, governments, and citizens. These alerts will provide warnings about high-risk areas and the probability of a flooding event. This, in turn will bring a range of benefits for the local council, for example, if we can predict where and when it will flood. Then traffic routes can be optimised, infrastructure can be optimised and refined, and nature can be preserved.

Another key aim is to support environmental research and education by providing access to local climate data, which can be used by university staff, students, and city planners for further analysis and modelling.

The thing that sets this apart from the existing weather stations and businesses is how we are focused on local environmental data across multiple sites rather than just relying on a singular, centralised site such as the Footscray weather station. While there are other options which offer some of the required

functionality, they all have limitations, such as being costly, too focused on consumers, and lacking durability and precision.

In contrast, our solution is designed to balance cost effectiveness, scalability, and integration with AI based systems. By focusing on easily deployable sensors with open data capabilities, this project can fill a gap in the current market: the lack of a flexible, council friendly weather monitoring solution which can operate across multiple small areas, and feed into a larger system. This creates an operational advantage for stormwater management and also opens possibilities for collaboration.

Purpose and Scope

The purpose of this smart drainage system project is to make the Maribyrnong City Council's original efforts surrounding their stormwater infrastructure management plans to become more impactful. By developing a sustainable forecasting model that leverages weather data collected from the council's network of weather stations and including third party sources such as the Bureau of Meteorology, Environment Protection Authority, and Melbourne water, our project can be seamlessly integrated into the already developed system by adding features that allow for users to be informed and given a sense of control over how they proceed during their day by providing early warnings for extreme weather events, specifically flood and overflow risks.

Our vision for this smart drainage system is to firstly outline in strict yet brief details what needs to be addressed and what can be done to mitigate or solve these issues in sight of economical limitations. By first understanding the systems and infrastructure already set in place, we can minimise the amount of economic stretch or creep that could occur with our proposal. To successfully implement this project, we illustrate to the client the successful completion of a quantifiable impact on system performance, cost and time. By creating a predictive model, our team, with the permission of the client, can translate raw numerical and non-numerical data into tangible information that can be used to send out warnings to users of this network.

Our scope consists of:

- Collection and storage of weather-related data
- Data mining techniques to develop a predictive model that can recognise patterns and forecast flood risks
- Cross referencing and comparison of interpolated data to improve model accuracy
- Visualisation of data using visualisation tools to illustrate weather trends and understand risk thresholds.
- Providing feedback to the council and communicating possible recommendations on optimal sensor placement to improve data coverage and monitoring accuracy

Functional Requirements

Environmental Sensor Integration

- Data gathered from localised weather stations should include relevant attributes such as precipitation, wind, humidity, temperature, as well as geolocation and timestamps for location-based analysis.
- The system must be able to fetch reference data from the EPA Footscray station and/or other third-party weather APIs (e.g., Bureau of Meteorology) for unknown data.
- Incoming data streams collected in real-time from localised weather stations must align and synchronise with data collected from the deployed storm water sensors.

Data Preprocessing

- Methodologies will be set for cleaning and preprocessing data to handle missing values, standardise formats, and trim unnecessary data points (e.g., outliers caused by faulty sensor readings such as negative rainfall).
- Pre-processed data must be validated and version controlled before it is used for training.
- Similar data types ingested from multiple sources should be converted into a common schema and use similar units.

Data Storage and Management

- A structured database should be used as a data store for collected data such as AWS S3 buckets.
- Sensor metadata such as device ID, geo tags, and system logs must also be stored persistently.
- Automatic backups of long-term storage must be performed daily to a secondary storage location for data redundancy.
- A data versioning system must be implemented to track changes over time, especially for datasets used in model training and testing. This should easily allow for rollbacks to changes made to datasets if errors are detected.

Predictive Analytics

- A machine learning model must be implemented that is able to predict potential overflow events that can trigger flooding in key areas.
- Historical data gathered from local weather stations must be used for correlation with historical flooding events, flood warnings, and drainage alerts.
- The resulting model should be able to classify associated overflow risks as low, medium, or high.

Data Visualisation and Reporting

- Localised weather station sensor readings should be easily understandable from the Maribyrnong Council's public data platform (Ubidots) so that comparison can be made with other open data platforms such as BoM's historical weather data.
- Dashboard systems should allow visualisation of data over specified time ranges (e.g., 24 hours, week, month). The dashboard system will support interactive visualisation of weather and flood risk data over a selectable time range using a combination of line graphs

to display rainfall, temperature, and flood risk thresholds, bar charts to show the frequency of sensor alerts, and where necessary, heatmaps to indicate flood risk locations, colour-coded to emphasise risk level severity.

- The system must support real-time monitoring of live weather data so that urgent notifications may be issued when overflow thresholds are reached and there is a risk of flooding.
- When thresholds are met, there needs to be visual representations to convey the urgency so that notifications may be broadcasted. These predicted flood risks may be represented visually via traffic-light colours to differentiate between low, medium, and high risks.
- The system must support customisable weather-based alert triggers, in addition to water-level thresholds. These triggers will monitor real-time sensor data for specific conditions such as:
 - when rainfall exceeds X mm/hr
 - when wind speed reaches Y km/hr
 - when humidity rises above $Z\%$

When a trigger condition is met, the system must:

- ✓ log the event with a timestamp
- ✓ send an automated alert (email or SMS) to designated council staff
- ✓ highlight affected locations on the dashboard in a distinct alert colour based on severity
- Threshold values should be configurable through the admin panel, allowing council staff to adjust them based on seasonal or environmental needs.

Alerting and Notification Systems

- Notification systems must be automated and should alert the Maribyrnong City Council when likelihood of overflows or flooding reaches pre-defined thresholds. This system facilitates the transition from passive monitoring to proactive risk management.
- Notification systems must be capable of broadcasting alerts through multiple channels such as in-app dashboard, email, SMS (configured via third-party), mobile push notifications, webhooks or APIs.
- Broadcasted alerts must leave a trail of event logs for auditing that include timestamps, alert types (pre-defined thresholds or predictive), and intended recipients.
- Alerts must follow standardised templates to easily automate and streamline the notification process and to ensure that alerts are consistent, clear, and actionable. Template structure can include the following components:

🚨 Flood Risk Alert	
Location:	VU Footscray Park Campus
Predicted Risk Score:	0.92 (Very High)
Expected Timeframe:	Next 2-4 hours
Rainfall Forecast:	38mm/hr
Action:	Inspect nearby drainage sensors.

- The alert system must support a test mode to simulate the sending of alerts without sending actual notifications.

Non-functional Requirements



System reliability

The system must always be reliable for data transmission, ingestion, storage, and calculations. During data transmission in sensor integration, the system should have a maximum latency of 2 seconds. Data should be gathered in a fast-paced manner, in addition an algorithm needs to be set to always check if the sensors have received the most recent available weather data to ensure that devices are synchronised. In addition, there should be identifiers to every data packet to check for data duplication.

In data preprocessing, performance should not be compromised when dealing with large datasets. Performance is an important factor in preprocessing the data and regulating the missing values. Any outliers should be flagged. In terms of data storage, the system requires backup functionality to recover the data in case of data loss.

Uptime monitoring

In data storage management, the database must be running, accessible, and always operational. The availability of the database is a cornerstone to the system functionality. The model trained to implement predictive analytics must always be available with no downtime. This is because as the new weather data is ingested into the sensor, the model should be ready to predict the flood risk on-demand and without delay. The same applies to data visualization, reporting and notification systems. All these systems require constant uptime and plans in place for troubleshooting issues when downtime occurs.

System redundancy

System redundancy is the fault tolerance of a system. For a system that aims to predict possible flooding, it is imperative to have fault tolerance set up to replace failed components and keep operations smooth. Fault tolerance is key when data storage and management are discussed. Here is what redundancy would look like in the sensors: a sensor might malfunction when reading data or transmitting other parameters, in this case an extra sensor is configured to be on standby and take over when the primary sensor fails.

API support

The sensor integration system uses API to get data from third parties. It is necessary that the API is secure. According to the Australian Government (n.d), to ensure API security, all traffic must use HTTPS, all certificates must be from SHA-2 with minimum key length of 2048, reject HTTP traffic, unused HTTP methods should be disabled and return HTTP 405. These common security practice must be in place to ensure that malicious parties do not access the sensor. In addition, notification APIs will be used for push notifications. The notification should arrive within 10 seconds of the report being finalized, therefore the TTL is 10 second and if it has surpassed 10 seconds then a new notification is to be sent out.

Scalability

When ingesting data into sensors, it is necessary that the sensor can handle a heavy load of data and be scalable when working with it. To resolve the issue of scalability, it always helps to use different types of data processing. Stream processing should be used for real-time data ingestion and for batch processing for non-urgent tasks.

Use Case

Extreme Weather Event Notification

- The system sends out a real-time alert to council officers and residents based on the predictive model output.
- Primary Actor: Council Officer – receives and acts upon flood alerts
- Secondary Actors: Residents (public user) - receive alerts to take precautions, AI Flood Prediction System – evaluates live data and triggers alerts, Notification System – responsible for formatting and delivering alerts.
- Preconditions: Sensors, An AI model, Alert Thresholds, Communication channels
- Basic Flow (Main Scenario)
 - i. The System collects real-time data from the environmental sensors
 - ii. The AI model processes this data and evaluates risks
 - iii. Risk prediction exceeds the alert threshold
 - iv. The alert model is flooded with details (Location, Risk level, Forecast)
 - v. Said Alert is framed and distributed into the: Internal System – Council Dashboard, Push Notification – Public App, Email and SMS gateways
 - vi. Alert is logged in the event history with timestamp and delivery method
 - vii. Council Officer receives the alert and initiates an internal response plan
 - viii. Residents receive alerts and can make safety decisions (avoid specific travel routes, securing and preparing private property)
- Alternative Flows
 - No Flood Risk Detected: No Alert is sent
 - Communication Channel Fails: If SMS fails, retry and use email or push notifications through an app.
 - AI System Delay: If real-time predictions are delayed, the alert would be flagged as “Delayed Prediction” including the timestamp in which the prediction is made along with a warning about “This is not real-time”
 - Duplicate Alert: if another identical alert is already active, the system updates the alert with the newest timestamp instead of transmitting a brand-new alert
- Postconditions
 - The alert is successfully received and acknowledged by the recipients
 - The alert status is saved for future auditing and reporting
 - Council begins emergency procedures if required
- Exceptions
 - Sensor malfunction
 - API downtime
 - User opt-out

Generating Risk Report

- The system allows a council officer to generate and download a report that summarizes environmental conditions, flood risk assessments, and alert histories for a specific time and data range. This report can be used for strategic planning, emergency response, community transparency, and system evaluation.
- Actors
 - Primary – Council officer who views and initiates a flood assessment risk report and overlooks system performance

- Secondary – AI Prediction System, Sensor Data Management System and Reporting Module
 - The Prediction System provides a risk score and evaluates historical data.
 - The Sensor Data Management System supplies raw data then translates the data for the initial report
 - The reporting module compiles the data and translates it into a readable format
- Preconditions
 - The sensor data has been collected and stored successfully for the reporting period
 - AI models have processed the historical data and generated risk scores
 - The user has been verified and authenticated to be granted report generation permissions
 - The system's reporting services are working and functioning
- Basic Flow
 - Officer logs into the system dashboard
 - Officer navigates to the "Reports" and/or "Flood Risk Report" sections
 - Officer selects a period of time
 - System fetches:
 - Environmental sensor data
 - AI-generated flood risk scores
 - Alert logs
 - The reporting module compiles data into a structured report:
 - Summary
 - Risk Trends
 - Timeline of high-risk events and alerts issued
 - Sensor performance and system uptime
 - The report is visualized in the dashboard view (charts, tables, graphs etc)
 - The Officer clicks "Download Report" to export the report as a PDF or CSV file
 - The report is archived in the system's document repository
 - The Officer may choose to share, print or display said report for planning or presentation purposes
- Alternate Flows
 - No data available for a period
 - Partial data available
 - Report generation delay
- Postconditions
 - Flood risk report is generated, downloaded and/or stored
 - The council has access to an evidence-based, scientific summary of flood risk conditions for review, policy and response planning
 - A log of the generated report is saved in the system
- Exceptions
 - Report generator crash
 - Corrupted data points
 - Permission errors
- Notes and Issues
 - Ensure the report is compliant with the Maribyrnong Council's formatting standards
 - Reports should include both technical insights and layman summaries for public communication

Configuring Alert Thresholds

- The system administrators can configure, update or overhaul the threshold levels for weather conditions and flood risks that will trigger alerts. These thresholds ensure that the AI model and notification system only act when the conditions surpass the defined parameters. This would ultimately prevent false alarms.
- Actors
 - Primary Actor:
 - System Admin – this person updates and saves the alert threshold values for the prediction system
 - Secondary Actor:
 - Flood Risk Prediction Engine – uses these thresholds to determine when to generate alerts
 - Notification Module – the module references the thresholds to determine when and how to broadcast alerts
- Preconditions
 - Administrator is logged into the system with the correct permissions
 - Prediction model and sensor network are operational
 - Existing threshold values are available for editing/reviewing
 - Network and Database connections are stable
- Basic Flow
 - Admin logs into the management dashboard
 - Admin navigates to the “Alert Settings” or “Threshold Configuration” section
 - System displays current thresholds:
 - Rainfall
 - Windspeed
 - AI flood risk score (e.g {>0.85 = High})
 - Drainage sensor water level
 - Admin selects a threshold to modify
 - Inputs new threshold values
 - Clicks “Save” and applies the new configuration
 - System validates input against set parameters and confirms success
 - New thresholds are stored in system settings and immediately used by prediction engine
 - An entry is logged with the time, date, user and brief details of the changes made
- Alternative Flows
 - Invalid Input Detected
 - Permission Error
 - Threshold Already Active
- Postconditions
 - The new threshold settings are saved and become active
 - AI model and alerting engine immediately use the updated values for risk evaluation
 - A log of the update is recorded in audit history
- Exceptions
 - System Timeout or Server Error
 - Threshold Conflict
 - Corrupted Configuration File
- Notes and Issues

- Threshold tuning should be based on analysis of historical data and community safety policies
- Consider adding a “Restore to Default” option for quick rollback
- Changes to thresholds should trigger a notification to all admins for transparency
- In future versions, support time-based thresholds during stormy seasons

Deploying Environmental Sensor

- Brief Description: Technician installs and configures a weather or water sensor in the field
- Actors
 - Primary Actor: Field Technician
 - Secondary Actor: System Admin
- Preconditions
 - Location is selected and approved
 - Equipment ready
- Basic Flow
 - Technician installs sensor
 - Connects it to power and network
 - Sensor auto-registers with system
 - Admin confirms sensor is transmitting data
- Alternative Flows
 - Sensor fails to register – troubleshoot or replacement would be needed
 - No network – fallback to offline mode
- Postconditions
 - Sensor appears on the dashboard and streams the data successfully
- Exceptions
 - Faulty sensor hardware
- Notes and Issues
 - Calibration needed post-installation

Resource Management



Introduction

Effective resource utilisation is central to any project's successful outcome. It ensures that all inputs, whether people, materials, finances, or time, are used strategically and efficiently. By identifying and organising every necessary resource, risks can be mitigated, delays can be avoided, and we can deliver a higher quality functioning flood monitoring system within the provided timeframe and budget. This resource planning will allow us to respond flexibly to challenges and maintain momentum across every phase of this project.

Identification of Required Resources

Human Resources

- Project Manager: Oversees planning, progress tracking, risk management and team coordination
- AI developer: Designs, trains, and deploys the machine learning models
- Hardware Engineer: Assembles, tests, and deploys sensors
- Communication Lead: Prepares presentations and writes reports
- Unit Coordinator: Oversee the project and provides feedback
- Council: Provides information and insight regarding the project
- Lecturer: Provides academic guidance and support

Materials and Supplies

- Sensors
 - Water level sensors
 - Temperature and humidity sensors
- Electronics
- Laptops
- Power supply
- Battery
- Mounting hardware
- Tripods or poles
- Weatherproof sensor housing
- Toolkits
- Screwdrivers
- Multimeters

Technology Resources

- Software
 - Python libraries
 - Web development frameworks
 - APIs
- Cloud services
 - AWS S3
 - phpMyAdmin
- Collaboration tools
 - Microsoft Office 365
 - Discord
 - Victoria University Community Campus

- GitHub

Time Resources

Throughout NEF3001

- Week 1: Choosing project and completing the Opening Sections
- Week 2: Inspecting sensors, communicating through meetings and working through the project proposal
- Week 3: Finalising the project proposal and preparing for the presentation
- Week 4: Presenting our project proposal

After NEF3001

- Week 5:
 - Set up essential services such as cloud and software
 - Start AI model development
 - Begin web/dashboard prototype display
- Week 6-7:
 - Integrate AI model with live sensor data
 - Conduct field tests in controlled environments
 - Begin stakeholder communication
- Week 8-9:
 - Develop visualisations and dashboard UI improvements
 - Optimize AI model performance and reduce false positives
 - Collect feedback
- Week 10-11:
 - Monitor system performance: uptime, accuracy, stability
 - Perform data validation and generate reports
 - Make any final tweaks to AI or backend systems
 - Practice team presentation and finalise documentation
- Week 12 onwards:
 - Conduct final project review
 - Finalise and submit report, dashboard, and system demo
 - Submit reflections and lessons learnt
 - Present to class and stakeholders

Legal and Ethical Resources

- Local council permits: for sensor placement in public spaces
- Risk assessment plan: for deploying sensors in outdoor environments

Knowledge and learning resources

- Academic journals/datasets: to compare performance and validate the AI model
- Case studies: past flood monitoring projects to learn from
- Training materials: tutorials, documents, or guides on frameworks

Allocation of Resources

-  – Materials and supplies
-  – Time resources
-  – Technology resources
-  – Legal, ethical, and academic resources

Resource Scheduling

The Gantt chart for the project's timeline shows when and how human resources are allocated – refer to the Timeline section of this Proposal.

Contingency Planning

To mitigate the potential risks, and ensure this project continues along its desired timeline, we have developed a robust contingency plan which is tailored to each area which poses a threat. These plans are designed to minimize disruption, reduce delays, and provide flexible responses to unexpected challenges.

Human Resources

- We will implement cross training where every team member has knowledge on each area of the project, meaning team members can step in for others if needed. Each task will have at least one back up person, so in case someone is unavailable, the task can still be completed.
- Weekly progress updates and shared documentation will ensure that work can be resumed without knowledge gaps. Progress updates will be completed through discord messages and calls, while shared documentation will be completed through office 365 using tools such as comments.

Materials and Equipment contingency

- We will ensure that all sensors and hardware are waterproof and placed in safe areas to ensure that there reduced risk of hardware failure

Technology Contingency

- We will ensure that there are regular backups of code and data in the cloud to ensure that we always have the latest backups
- To ensure uptime, the backend system will have local data caching so that sensor data can be stored for short amounts of time when there are network outages

Time Contingency

- The project timeline will have buffer weeks to account for unforeseen delays
- Tasks will be broken down into smaller steps so that they can be executed independently so that there is a reduced delay in one area
- If some major milestones are at risk, we will refocus on core deliverables to ensure a successful submission

Legal and stakeholder contingency

- We will have alternate deployment locations for the sensors in case public permissions are delayed or denied
- We have access to open data sources in case stakeholders are unable to provide confidential data
- Regular engagement with stakeholders is conducted to ensure expectations and deliverables are clear

Monitoring

Weekly team check-ins

- Regular meetings will be held every week to review task progress, identify blockers and reallocate resources if necessary
- Each team member will provide updates aligned with their assigned tasks, ensuring transparency and accountability.

Task Tracking

- Deadlines will be tracked with colour-coded priorities so that we can flag at risk activities to ensure activities are completed in a hierarchy that benefits the project
- We will use notion with a kanban board so that we can assign, monitor and update task statuses
- Each task will be categorized into one of five phases: Planning, development, testing, deployment, and community engagement. These phases will help us keep on top of where each task is at.

Hardware and infrastructure

- Sensor uptime, and data transmission will be monitored through a live dashboard
- Error logs and data inconsistencies will trigger alerts so that we can respond quickly
- Deployment sites will be inspected to check for any physical damage, vandalism or environmental wear.

Evaluation

Key performance indicators

- Will compare the sensor accuracy across different timeframes and compare with other sensors in a similar area to check if the sensor readings are accurate
- Measure the uptime of the system and see how long it can operate without interruptions
- Evaluate the percentage of planned tasks completed on time and within scope

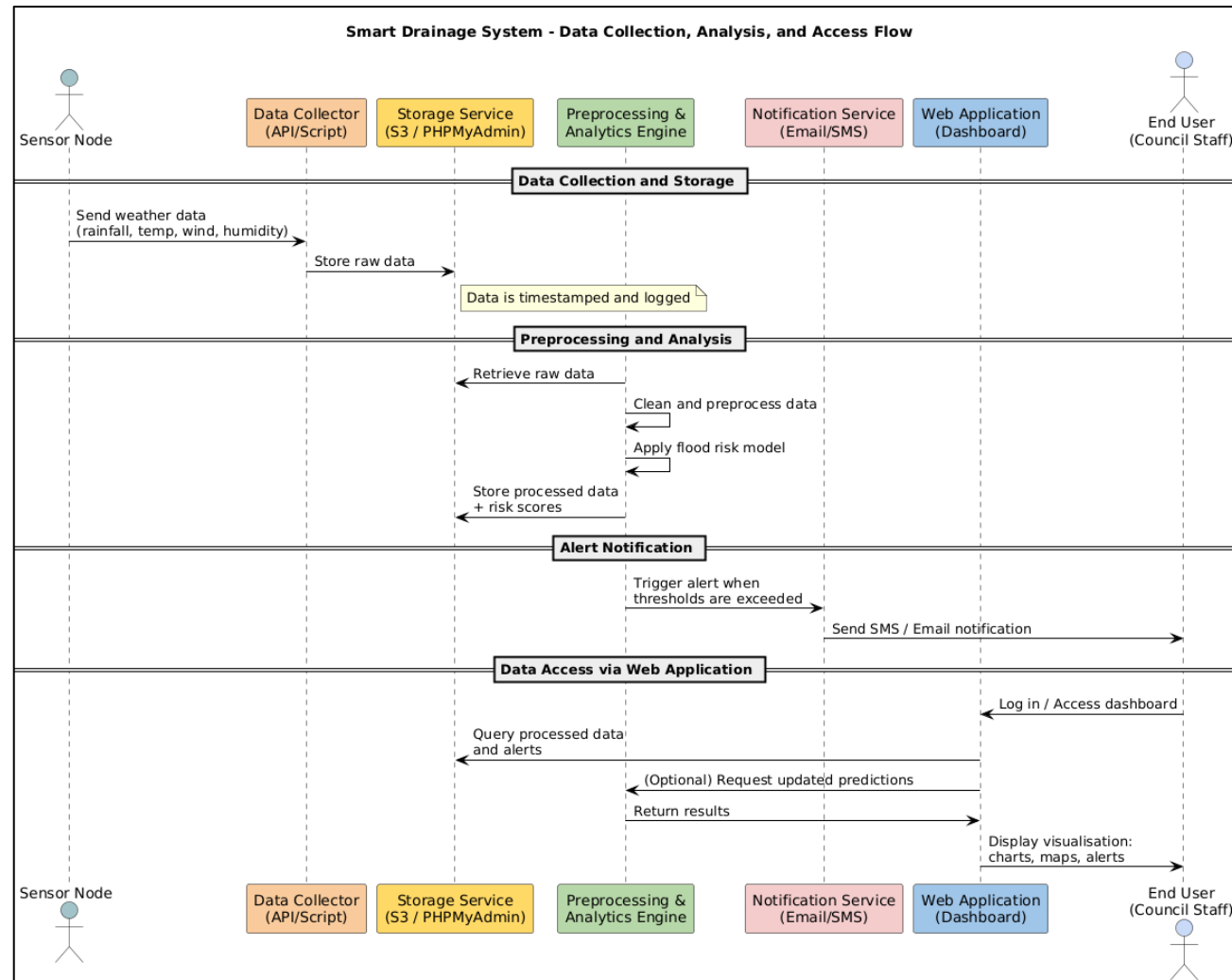
User and stakeholder feedback

- Will gather feedback from local councils so that we can evaluate the usefulness, clarity and ease of use
- This feedback will give us the opportunity to make changes which are suited towards the council needs

Post-Deployment evaluation

- A project review will be conducted at the end of the deployment phase
- We will analyse what worked well, what challenges emerged and what could be improved
- Lessons learnt will be documented and shared

Sequence Diagram



UML Source Code

```
@startuml
title Smart Drainage System - Data Collection, Analysis, and Access Flow

' Skin parameters
skinparam participantBorderColor Black
skinparam participantBackgroundColor White
skinparam sequence {
    ArrowColor Black
    LifeLineBorderColor Black
    LifeLineBackgroundColor #FDFDFD
    Participant {
        BackgroundColor #D0E6A5
        FontColor Black
    }
}

actor "Sensor Node" as Sensor #A2C4C9
participant "Data Collector\n(API/Script)" as Collector #F9CB9C
participant "Storage Service\n(S3 / PHPMyAdmin)" as Storage #FFD966
participant "Preprocessing &\nAnalytics Engine" as Engine #B6D7A8
participant "Notification Service\n(Email/SMS)" as Notifier #F4CCCC
participant "Web Application\n(Dashboard)" as WebApp #9FC5E8
actor "End User\n(Council Staff)" as User #C9DAF8

== Data Collection and Storage ==
Sensor -> Collector #DarkGreen : Send weather data\n(rainfall, temp, wind, humidity)
Collector -> Storage #DarkOrange : Store raw data

note right of Storage
Data is timestamped and logged
end note

== Preprocessing and Analysis ==
Engine -> Storage #Blue : Retrieve raw data
Engine -> Engine #Black : Clean and preprocess data
Engine -> Engine #Black : Apply flood risk model
Engine -> Storage #Teal : Store processed data\n+ risk scores

== Alert Notification ==
Engine -> Notifier #Red : Trigger alert when\nthresholds are exceeded
```

```
Notifier -> User #Red : Send SMS / Email notification

== Data Access via Web Application ==
User -> WebApp #DarkBlue : Log in / Access dashboard
WebApp -> Storage #Purple : Query processed data\nand alerts
WebApp -> Engine #Gray : (Optional) Request updated predictions
Engine -> WebApp #Gray : Return results
WebApp -> User #DarkRed : Display visualisation:\ncharts, maps, alerts

@enduml
```



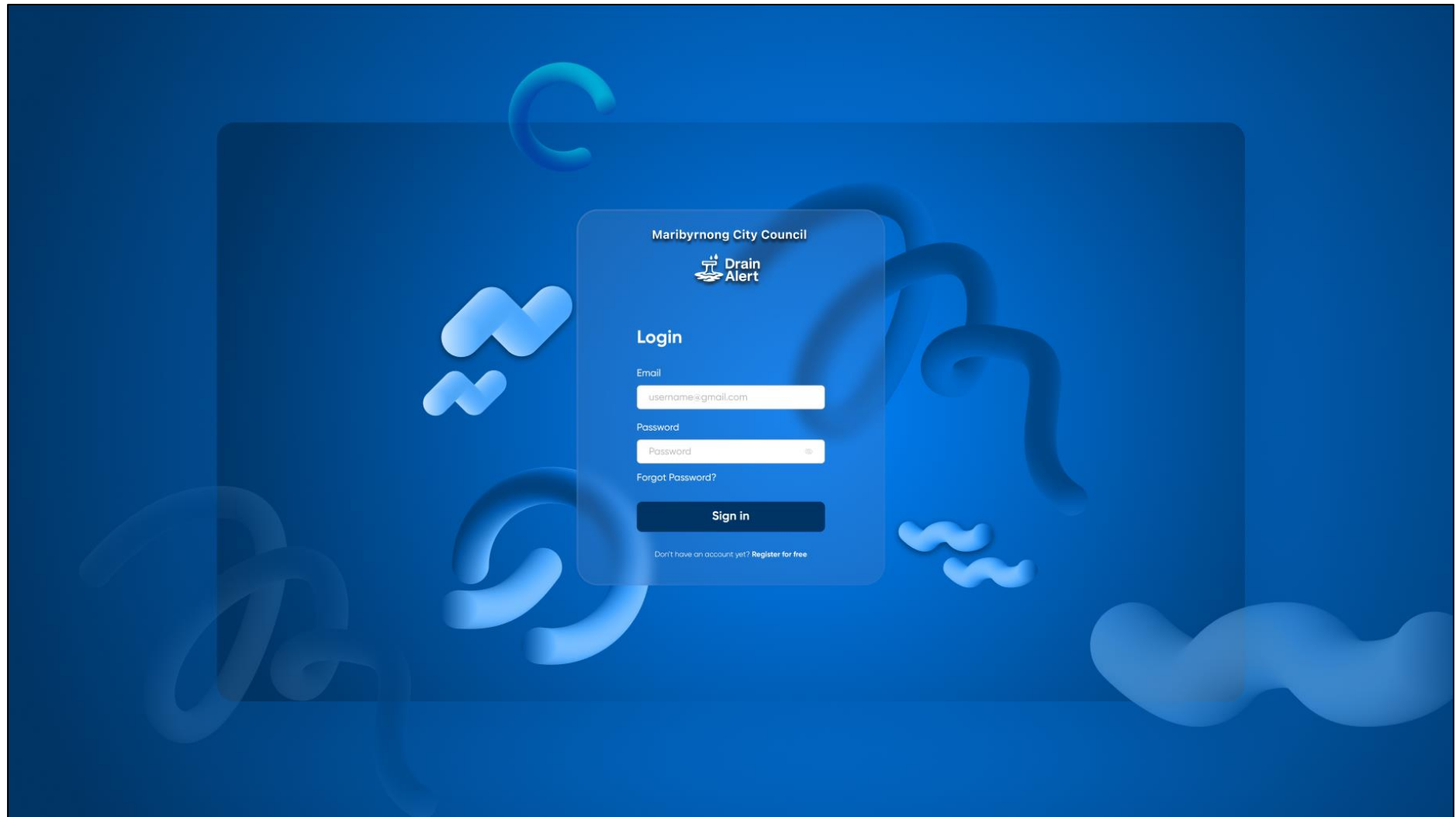
```
        "Risk Level: HIGH\n" +  
        "Recommended Action: Dispatch maintenance team."  
  
        CALL sendEmail(to = "council.staff@maribyrnong.vic.gov.au", subject =  
"High Flood Risk Alert", body = message)  
        CALL sendSMS(to = "+61XXXXXXX", body = message)  
  
    ELSE:  
        CONTINUE
```

Data Query via Web Application Dashboard

```
INPUT user_query_parameters  
FETCH sensor_data FROM database WHERE  
    location == user_query_parameters.location AND  
    timestamp BETWEEN start_time AND end_time  
DISPLAY sensor_data ON dashboard
```

The pseudocode provided above reflects a conceptual design. Any specifics necessary for project implementation such as database schema, API integration and communication protocols, and potential machine learning integration will be defined once access to data becomes available.

Login Page



Maribyrnong City Council

 Drain Alert

Login

Email

Password

[Forgot Password?](#)

[Sign in](#)

Don't have an account yet? [Register for free](#)

Risk Management Plan



Risk Assessment

RISK ID	DESCRIPTION	LIKELIHOOD	IMPACT	RISK LEVEL
1	Data gathered from weather stations are missing timestamp values: this can cause some data anomalies in the prediction model. It will not understand when an event has occurred and fail to create a timeline. Our model requires timestamps for both predictive accuracy and data integrity.	Moderate	High	High
2	Connectivity loss due to extreme weather or signal obstructions: this can cause the Elsys ultrasonic sensor to lose connection with the main gateway and be unable to transmit vital data readings (e.g., temperature, water level, humidity, and other attributes).	Low	High	Moderate
3	Desynchronized clocks among systems: this can cause data mismatch and other incompatibility issues between systems.	Low	Moderate	Moderate
4	When preprocessing data, important information should not be omitted by mistake, null values must be handled the correct way. If a data value is trimmed unnecessarily, then the whole record may be rendered invalid and fail to provide meaning to the predictive model.	Low	Moderate	Moderate
5	Data inconsistencies. Data inconsistency means inconsistent measurement units in different attributes, resulting in inaccurate correlation and subsequently, faulty predictions.	Moderate	High	High
6	Web application bugs and errors may lead to the web application crashing and not showing information upon request.	High	High	High

7	One important non-technical risk to the project is if a team member leaves the group and their responsibilities remain unallocated, causing panic among the team.	Moderate	High	High
8	Running behind schedule and being unable to deliver the project on time for different reasons such as illness, budget constraints, or a team member leaving prior to project deployment.	Low	Moderate	Moderate
9	Misconfigured thresholds and attributes in the notification system can cause inaccurate readings, incorrectly triggering risk level indicators. This type of error can result in false positive or false negative notifications being sent to the council.	Low	High	Moderate
10	Failure to meet regulatory compliance. When collecting environmental data, adherence to state laws regarding data collection and data privacy must be a priority. Non-compliance may result in lawsuits.	High	High	High

Risk Mitigation

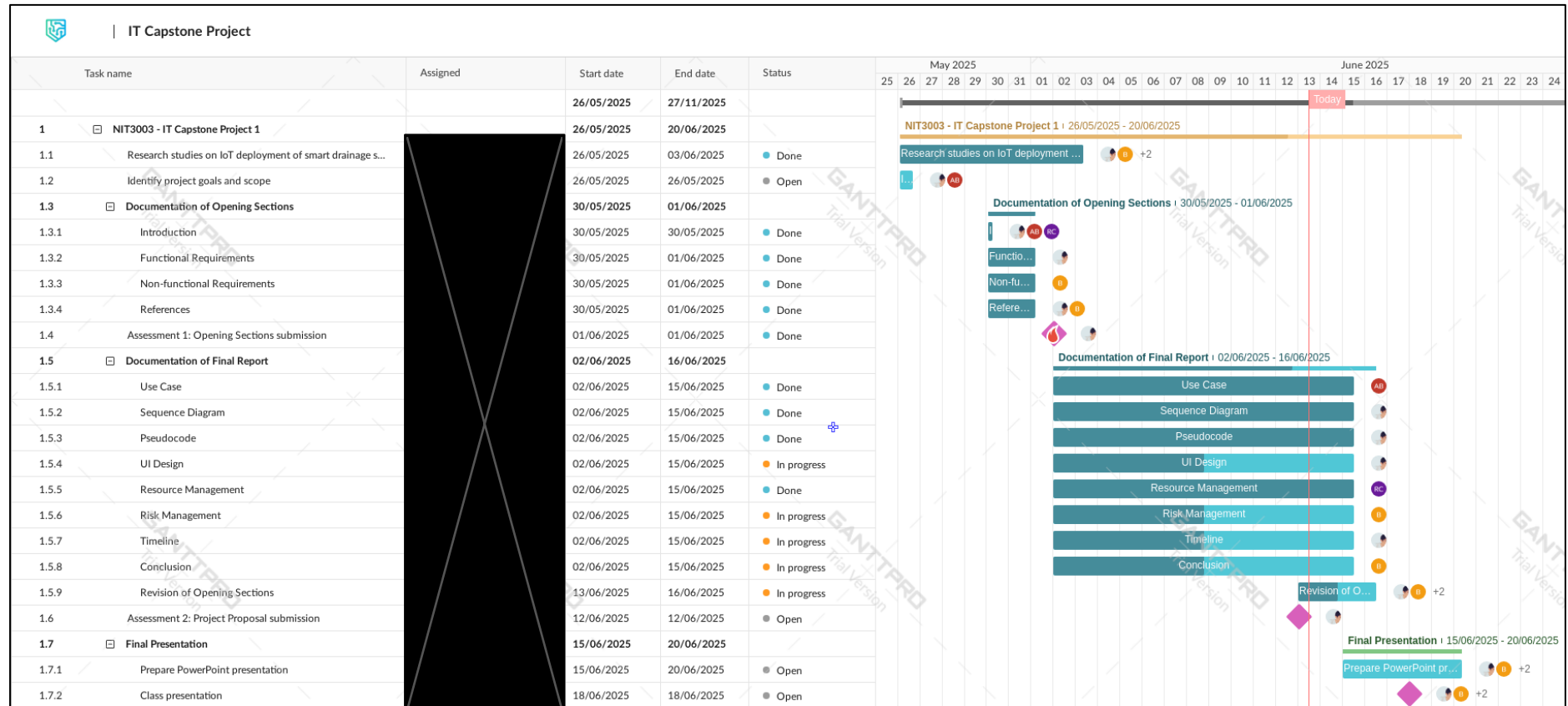
RISK ID	MITIGATION STRATEGY	OWNER
1	To mitigate this risk, data must be sourced from a reputable provider that offers reliable and accurately timestamped data points. It is recommended prioritising weather data provided by the Bureau of Meteorology over commercial stations. Do not use data with daily timestamps—hourly or more frequent intervals are required.	Data analyst

2	To prevent connection loss, it is important to place the sensor in a place and at an angle where the antenna is not obstructed by any object to ensure reliability during data transmission.	Technician
3	Synchronise the clocks between participating systems and devices, and make sure they operate using the same standards.	Technician
4	Use preprocessing best practices, according to Abrahams (2023). Following data preprocessing best practices improves accuracy, gives better insights and allows for better model performance.	Data analyst
5	As suggested in the functional requirements, all attributes must use the same schema when training the predictive model.	Data analyst
6	Create a refresh or reload button that would allow the user to re-execute the backend code, causing the information fields to be repopulated or re-rendered on the interface.	Front-end developer
7	In the event that a team member leaves or becomes unavailable for an extended period, their workload must be reallocated among the remaining team members to ensure continued progress.	Team leader
8	If the project is running behind schedule, then the team leader must formally notify the council via letter or email advising them of the circumstances and requesting for an extension. The letter must outline the reasons for the delay and provide a revised estimated completion date, contingent on the approval of the extension.	Team leader
9	To prevent threshold misconfigurations, it is important that the data analyst provide the accurate threshold values derived from the data analysis. This ensures that notifications are correctly triggered and sent to the council based on the assessed flood risk levels.	Data analyst & back-end developer
10	Strict adherence to Victoria's environmental data collection regulations is essential when collecting data for the project. It is the team leader's responsibility to review relevant legislative requirements and ensure full compliance throughout the data collection process.	Team leader

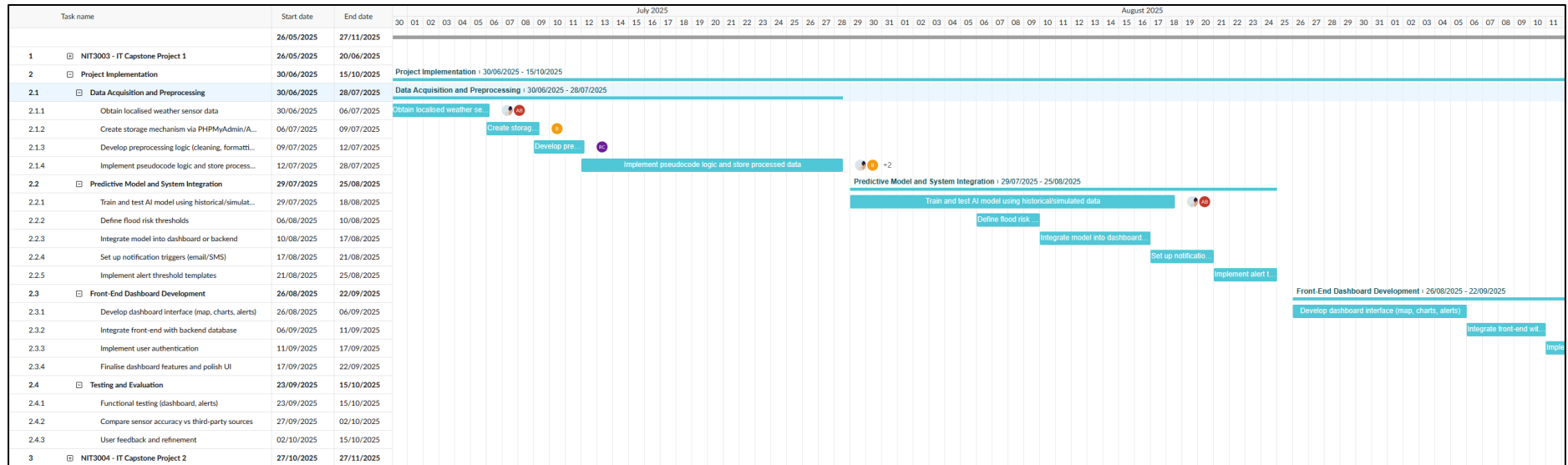
Timeline

Note! Please find attached with this Project Proposal a high-resolution copy of the project timeline.

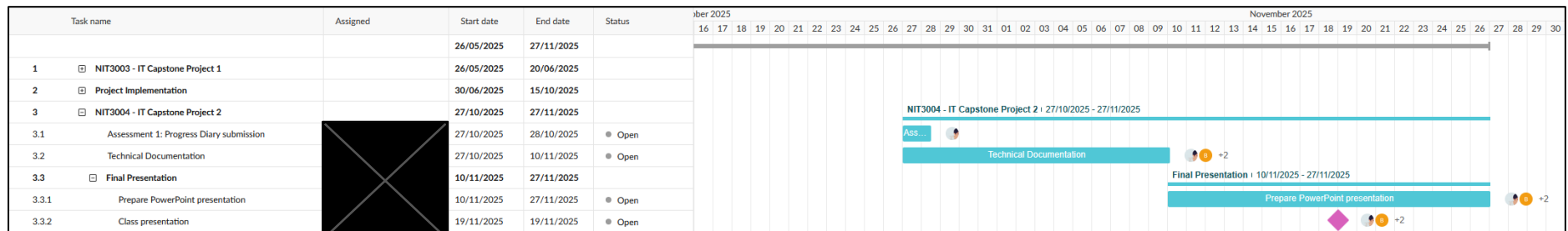
NIT3003 – IT Capstone Project 1



Project Implementation



NIT3004 – IT Capstone Project 2



Conclusion



In conclusion, this project presents an innovative and smart approach to flood prediction utilising data mining techniques and data analysis. By implementing a predictive model trained by historical data, taking into consideration the various attributes such as water levels, rainfall, temperature and humidity, the model should offer highly accurate flood risk assessments. The Council plans to integrate this predictive model into their existing drainage system to enhance flood management.

The integration of this predictive model into the drainage system should allow the council to monitor for flood risk in real-time. Real-time data from sensors and weather stations can feed into the model and predict flood risks instantly.

Additionally, the project promotes community awareness by allowing web application access for the council members to observe latest flood risk developments in the Maribyrnong area.

To summarize, by combining innovative technologies and data analysis techniques this project will deliver accurate flood prediction, positively impact emergency response times, and protect the community from unexpected overflow and flood events.

References

- Sitzenfrei, R., Annus, I., Langeveld, J., Rieckermann, J., & Rauch, W. (2024). Developments and applications of IoT-based sensors for wastewater and drainage systems. *Water Science & Technology*, 89(4), iii-v. <https://doi.org/10.2166/wst.2024.058>
- Zidaouim, I., Wemmert, C., Dufresne, M., Joannis, C., Isel, S., Wertel, J., & Vazquez J. (2023). Validation of wastewater data using artificial intelligence tools and the evaluation of their performance regarding annotator agreement. *Water Science & Technology*, 87(4), 2957-2970. <https://doi.org/10.2166/wst.2023.174>
- Kiruthika, E., Kapilan, M., Barathraj, S.K., Hariprakash, P., & Divraj, M. (2025). IoT-Enabled Smart Drainage Monitoring System for Urban Infrastructure. *International Journal of Engineering Research in Computer Science and Engineering*, 12(4). <https://www.ijercse.com/article/1%20April%202025%20IJERCSE.pdf>
- Singh, A., Tandel, N., Baria, K., Kamli, K., & Kannan, S. (2020). Smart Drainage Monitoring System with Weather Forecasting And Rain Water Harvesting. *International Journal of Creative Research Thoughts*, 8(4), 3115-3119. <http://www.ijcrt.org/papers/IJCRT2004441.pdf>
- Chandru, P., & Vadivel, R. (2024). IoT-based intelligent drainage and dust identification system for enhanced hygiene and efficiency. *World Journal of Advanced Research and Reviews*, 21(3), 1789-1797. <https://doi.org/10.30574/wjarr.2024.21.3.0683>
- Abdullah, H. (2023, May 10). Understanding PostgreSQL and its strong embrace of ACID properties. *Dev.to*. <https://dev.to/pawnsapprentice/understanding-postgresql-and-its-strong-embrace-of-acid-properties-2dpl>
- Russell, S., & Norvig, P. (2021). *Artificial intelligence: A modern approach* (4th ed.). Pearson.
- Bureau of Meteorology. (n.d.). *Flood warning services*. <http://www.bom.gov.au/water/floods/floodWarningServices.shtml>
- Maribyrnong City Council. (2016). *Flood Management Plan for Maribyrnong City Council and Melbourne Water: June 2016–June 2021*. <https://www.maribyrnong.vic.gov.au/files/assets/public/council-plans-reports-and-publications/emergency-management/flood-management-plan-2016-2021.pdf>
- Maribyrnong City Council. (2024). *Integrated Water Management Plan*. https://www.maribyrnong.vic.gov.au/files/assets/public/v/1/council-strategies-and-policies/maribyrnong-integrated-water-management-plan_v7_final.pdf
- Smart City Strategies & Solutions. (2025, March 3). *Smart City Strategies for Climate Resilience: Building Flood-Proof Australian Cities*. <https://www.smartcityss.com/resources/smart-city-strategies-for-climate-resilience-building-flood-proof-australian-cities>
- Australian Government. (n.d.). API security. *Australian Government*. <https://api.gov.au/sections/api-security.html>
- IMARC Group. (2024). *Australia Smart Water Management Market Size, Share, Trends and Forecast by Component, Application, and Region, 2025-2033*. <https://www.imarcgroup.com/australia-smart-water-management-market>

Abrahams, A. (2023, August 17). *Best practices for data cleaning and preprocessing*. Jumping Rivers.
<https://www.jumpingrivers.com/blog/best-practices-data-cleaning-r>